

Self-Fulfilling Inflation Risk Premia under the Fiscal Theory

Equilibrium Multiplicity and Government Debt Maturity

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Outline

1 Motivation

2 Model

3 Two Equilibria

4 Dynamics

5 Debt Maturity and QE

6 Discussion

The Puzzle: QE's Ambiguous Inflation Effects

Large-scale asset purchases, maturity operations, and fiscal-risk episodes have **ambiguous** effects on inflation expectations.

- Sometimes government debt behaves like a **safe asset**: higher debt valuation coincides with *lower* inflation compensation.
- Sometimes fiscal news raises inflation expectations and yields *together*.

Standard accounts: the difference is in fundamentals — fiscal position, policy regime, business cycle.

This paper: **the same fundamentals support both outcomes**, depending on whether investors have coordinated on pricing fiscal risk at all.

Result preview

A maturity shortening is **only half as disinflationary** when fiscal risk is priced — and reverses sign once the fiscal loading is large enough. Same parameters, rules, states.

Where Uniqueness Breaks

Standard FTPL (Leeper 1991; Cochrane 2001) — active fiscal + passive monetary pins P_t *uniquely*:

$$\frac{B_t}{P_t} = \mathbb{E}_t \sum_{j \geq 0} m_{t,t+j} s_{t+j}$$

B_t predetermined + surpluses tied to fundamentals $\Rightarrow P_t$ unique.

The hidden premise: that $m_{t,t+j}$ is itself determined by fundamentals.

But the SDF is an **equilibrium object**. It must price every risk bondholders bear, including inflation risk. So:

- Whether inflation responds to a fiscal-uncertainty shock depends on whether bond values load on it (A_2)
...
- ... which depends on whether the SDF prices it (μ) ...
- ... which depends on whether inflation responds to it.

This circularity has two self-consistent resolutions.

Positioning: What Exactly Is Indeterminate?

Literature	What is indeterminate	RP sign	Role of maturity
Self-fulfilling crises Calvo 88; Cole–Kehoe 00; Aguiar et al. 22	Default / rollover, via solvency	Positive	Secondary
Debt dilution Aguiar–Amador 20	Debt path, via beliefs about future pricing	Positive	<i>Necessary</i>
Safe-asset status Brunner–meier et al. 20, 22	Bubble / service-flow premium	Positive (vicious)	Secondary
Monetary indeterminacy Benhabib et al. 01; Davig–Leeper 06	Steady state / regime	Not priced	Absent
This paper	Whether fiscal uncertainty is priced <i>at all</i>	Negative (hedge)	Amplifies, not necessary

- **Multiplicity in *what is priced*, not in prices or paths.** Both equilibria: same fundamentals, same rules, finite PV of surpluses. They differ in *one* coefficient, the SDF loading μ .
- **Sign is the mirror image of the crisis literature.** There, believing debt is risky raises the burden and validates the belief. Here, believing fiscal risk is priced makes debt *more* valuable and validates *that*.

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Environment

Two orthogonal unit-normal shocks, $\text{Cov}(\varepsilon_{t+1}, \eta_{t+1}) = 0$; two states.

Real rate — Gaussian AR(1):

$$r_{t+1} = \bar{r} + \rho_r(r_t - \bar{r}) + \sigma \varepsilon_{t+1}$$

Fiscal uncertainty — CIR, preserves affine structure:

$$\xi_{t+1} = \bar{\xi} + \rho_{\xi}(\xi_t - \bar{\xi}) + v\sqrt{\xi_t} \eta_{t+1}$$

In what sense is η non-fundamental?

ξ_t enters *no* surplus rule, technology, preference, or policy rule. It appears nowhere except through the loading A_2 and the risk price μ .

Fundamental fiscal risk = actual uncertainty about surpluses.

Coordinated fiscal-risk pricing = whether that risk is in the SDF.

The multiplicity is about the *second*, not the first.

Government, Policy, and the Pricing Kernel

Geometric-coupon bonds $\omega \in [0, 1]$; real debt value $b_t = Q_t B_t$; flow budget constraint (*not* balanced budget):

$$\frac{b_{t+1} + s_t}{b_t} = \frac{r_{g,t+1}}{1 + \pi_{t+1}}, \quad s_{t+1} = s^* \left(\frac{b_t}{b^*} \right)^{\delta_b}$$

Log-linearized: $r_{s,t+1} = k_0 + k_1 \log b_{t+1} + (k_2 \delta_b - 1) \log b_t$.

Passive Taylor rule ($\rho_\pi < 1$): $i_t = i^* + \rho_\pi (\pi_t - \pi^*)$.

Affine SDF (conjectured, verified in equilibrium):

$$m_{t+1} = -\delta - r_t - \lambda \varepsilon_{t+1} - \mu \sqrt{\xi_t} \eta_{t+1}$$

NKPC closes the loop — if π responds to η , output does, and log utility forces the SDF to load on η :

$$\kappa y_t = (\kappa y_n - \psi_r r_t) + (\pi_t - \beta \mathbb{E}_t[\pi_{t+1}])$$

Duration Enters Through *One* Factor

An inflation surprise raises i via the Taylor rule, depressing the long bond's resale price.
Capitalizing the geometric stream:

$$r_{g,t+1} - \mathbb{E}_t[r_{g,t+1}] = -\gamma(\pi_{t+1} - \mathbb{E}_t[\pi_{t+1}]), \quad \gamma \equiv \frac{\omega\rho_\pi}{1 - \omega\rho_\pi}$$

Substituting into the budget-constraint innovation and solving:

$$\pi_{t+1} - \mathbb{E}_t[\pi_{t+1}] = -PT(r_{s,t+1} - \mathbb{E}_t[r_{s,t+1}]), \quad PT \equiv \frac{1}{1 + \gamma} = 1 - \omega\rho_\pi$$

- PT is the fraction of a surplus surprise that reaches the *price level* rather than being absorbed as a capital gain.
- $\omega = 0 \Rightarrow \gamma = 0, PT = 1$: the one-period Fisher identity.
- Baseline $\omega = 0.975, \rho_\pi = 0.85 \Rightarrow \gamma \approx 4.84, PT \approx 0.171$.
- Every ω -dependent object below is a one-period coefficient $\times PT$.

Risk Prices Are Equilibrium Objects

Log utility + market clearing: $m_{t+1} - \mathbb{E}_t[m_{t+1}] = -(y_{t+1} - \mathbb{E}_t[y_{t+1}])$.

Taking the unexpected NKPC and iterating the passive Taylor rule forward gives the forward-adjusted slope

$$\tilde{\kappa} \equiv \frac{\kappa}{1 - \beta\rho_\pi} \quad (\text{baseline } \tilde{\kappa} = 0.63 \text{ vs } \kappa = 0.10)$$

Matching shock loadings against the SDF:

$$\lambda = \frac{\sigma}{\kappa} \left[(1 - \beta\rho_\pi)B_2 + \beta - \psi_r \right] > 0, \quad \mu = \frac{\nu}{\kappa} \left[(1 - \beta\rho_\pi)B_4 - \beta(B_3 + B_4\rho_\xi) \right]$$

- $\lambda > 0$: the ψ_r term dominates — the bond is a deflation hedge.
- μ inherits the sign of $-A_2$: zero in Eq I, strictly *negative* whenever $A_2 > 0$.
- This uses the NKPC and ρ_π — neither is in the SDF by assumption.

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The Bond-Pricing Condition Is Quadratic in A_2

Conjecture $\log b_t = A_0 + A_1 r_t + A_2 \xi_t$. The Euler equation $\log \mathbb{E}_t[e^{m_{t+1} + r_{s,t+1}}] = 0$ gives three conditions; the ξ_t one is

$$A_2 \left(k_1 \rho_\xi + k_2 \delta_b - 1 + \frac{1}{2} k_1^2 A_2 v^2 - \mu k_1 v \right) = 0$$

Eq I — Fundamental

$A_2 = 0, \mu = 0$. Bond values load only on r_t ; fiscal uncertainty is *entirely irrelevant*.

Eq II — Fiscal Sunspot

With μ from above, $N \equiv 1 - k_1 \rho_\xi - k_2 \delta_b$:

$$A_2^* = \frac{N}{k_1^2 v^2 \left[\frac{1}{2} + \frac{(1 - \beta \rho_\pi) PT}{\kappa} + \frac{\beta N}{k_1 \kappa} \right]} > 0$$

Proposition

Under $\rho_\pi < 1$, $\delta_b < 1$, and $k_1 \rho_\xi + k_2 \delta_b < 1$, the FTPL bond-pricing equation admits exactly two RE equilibria *within the affine Markov class*: $(A_2, \mu) = (0, 0)$ and (A_2^*, μ^*) with $A_2^* > 0 > \mu^*$.

Within the conjectured class; does not rule out non-affine equilibria. The substantive point: even in the most disciplined class, the pricing of fiscal uncertainty is not pinned down by fundamentals.

The Self-Fulfilling Loop

belief $\longrightarrow$ pricing kernel $\longrightarrow$ bond value $\longrightarrow$ inflation $\longrightarrow$ belief confirmed

- 1 Investors coordinate on fiscal uncertainty being priced: $\mu < 0$.
- 2 Bond values rise with ξ_t : $A_2 > 0$. Debt is revalued upward whenever fiscal uncertainty rises — with *no* change in the surplus path.
- 3 FTPL: if b_t rises and surpluses are unchanged, the price level must *fall* on impact ($B_4 < 0$).
- 4 That is a low-inflation, low-output — hence *high-marginal-utility* — state.
- 5 A bond that gains value there is **insurance**: investors accept a lower expected return. $RP^\xi < 0$.
- 6 Inflation responding to η forces the SDF to load on η — **validating** $\mu < 0$.

In Eq I every link closes at zero. Both fixed points coexist under identical rules and states; only the coordination differs.

This Is *Not* a Default or Fiscal-Stress Channel

In the sovereign-debt literature, fiscal risk means higher yields, lower debt values, a *positive* premium. That intuition applies to a channel this model *does not* contain:

- No default, no solvency constraint, no rollover risk. The government always honours nominal obligations; the *only* adjustment margin is the price level.
- What is priced is the state in which FTPL revaluation forces P_t *down*.

Mirror image of the vicious circle:

Calvo 88 / Cole–Kehoe 00 / BMS 22	This paper
Believe debt is risky ⇒ demand compensation ⇒ raise fiscal burden ⇒ debt <i>is</i> riskier ($RP > 0$)	Believe fiscal risk is priced ⇒ bond becomes a hedge ⇒ debt revalued <i>up</i> ⇒ validates pricing ($RP < 0$)

A model with default, or with surpluses falling in the bad state, would deliver $RP^\xi > 0$ and reverse the loop. Excluded here by construction.

Calibration and Equilibrium Coefficients

Baseline (quarterly)

	Symbol	Value
Real rate	$\bar{r}, \rho_r, \sigma$	.01, .90, .005
Fiscal unc.	$\bar{\xi}, \rho_\xi, \nu$	.20, .96, .40
Fiscal rule	s^*, b^*, δ_b	.006, .60, .30
NK block	κ, ψ_r, β	.10, .50, .99
Monetary	ρ_π, i^*, π^*	.85, .03, .02
Duration	ω	0.975

$$D_{mac} \approx 18.7Q \approx 4.7\text{yr}$$

$$PT = 0.171, \gamma = 4.84$$

($\nu, \bar{\xi}, \rho_\xi$) set so the η shock is comparable to the ε shock and the 10y IRP in Eq I is $\approx -0.5\%/yr$.

Implied coefficients

Object	Eq I	Eq II
A_1 (debt on r_t)	-9.439	-9.439
A_2 (debt on ξ_t)	0	+0.206
λ (real-rate RP)	+0.037	+0.037
μ (fiscal RP)	0	-0.077
B_2 (π on r_{t+1})	+1.600	+1.600
B_4 (π on ξ_{t+1})	0	-0.035
B_1 (π on r_t)	-2.440	-2.440
$B_3 + B_4\rho_\xi$	0	+0.014
$B_1 + B_2\rho_r$ (Fisher)	-1.000	-1.000
RP^ξ/ξ_t	0	-0.006

Fisher consistency $B_1 + B_2\rho_r = -1$ holds to machine precision for all ω .

All entries generated by the solver (src/scripts/make_paper_tables.py), so the table cannot drift from the code.

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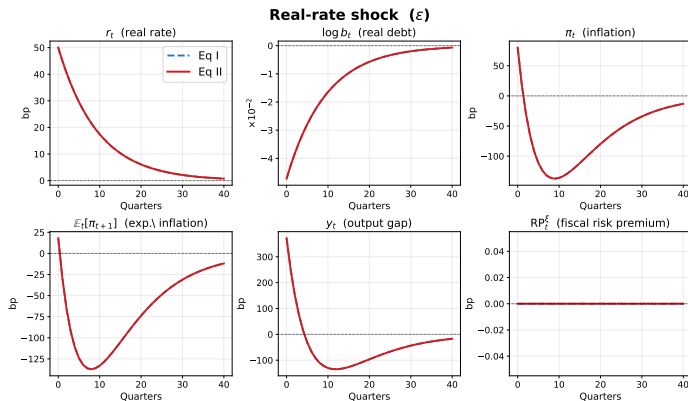
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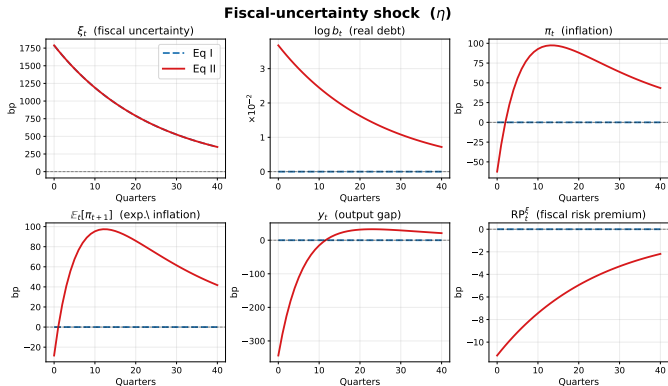
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Real-Rate Shock: the Two Equilibria Coincide



1 s.d. ε shock ($\sigma = 0.005$, 50 bp). $\log b$ falls $A_1\sigma \approx -0.047$; FTPL revaluation raises π on impact by $B_2\sigma \approx +80$ bp (stepping on a rake), then the backward channel ($B_1 < 0$) takes over, troughing near -137 bp at $t = 9$. ξ_t and IRP identically zero in both equilibria, by orthogonality.

Fiscal Sunspot Shock: the Central Result



1 s.d. η shock ($\nu\sqrt{\bar{\xi}} \approx 0.179$; ξ is normalized, $\bar{\xi} = 0.20$). Eq I (dashed): every panel except ξ_t is identically zero. Eq II (solid): $\log b$ revalued +0.037 on impact; impact *deflation* ≈ -62 bp ($B_4 < 0$); hump-shaped inflation peaking $\approx +97$ bp at $t = 13$; IRP falls ≈ 25 bp. Real rate unchanged throughout.

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Duration Acts Asymmetrically on the Two Channels

ω enters only through PT — but it enters the two channels *differently*:

Fundamental channel: *dampened*

PT multiplies the loading directly:

$$\frac{\partial B_2}{\partial \omega} = k_1 A_1 \rho_\pi = -7.94 < 0$$

Longer maturity absorbs surplus surprises as capital gains.

Sunspot channel: *amplified*

PT sits in the *denominator* of the fixed point:

$$\frac{\partial A_2^*}{\partial \omega} = +0.15 > 0$$

so $\partial(B_3 + B_4 \rho_\xi) / \partial \omega = +0.016 > 0$.

- Longer duration **dampens** fundamental inflation volatility while **deepening** the safe-haven IRP:
 $\partial \text{IRP} / \partial \omega < 0$.
- All closed forms verified against central finite differences of the solver to **max relative error** 8.1×10^{-11} .

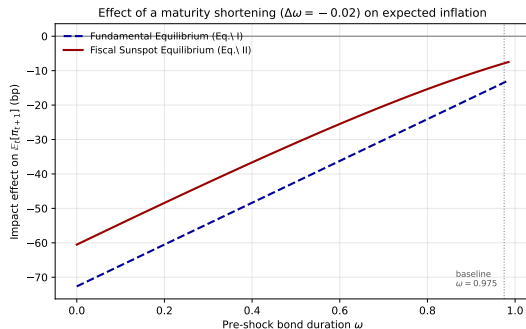
The Headline: How Much QE Disinflates Depends on the Equilibrium

MIT shock $\omega_H = 0.975 \rightarrow \omega_L = \omega_H - 0.02$, reverting at $\rho_\omega = 0.90$. Impact effect on $\mathbb{E}_t[\pi_{t+1}]$, in bp:

Channel	Source	Eq I	Eq II
Term-premium quantity	TP $\Delta\gamma$	+0.60	-0.19
Term-premium repricing	$\gamma \Delta TP$	-1.15	-0.68
A_0 balance-sheet hedge	$\lambda \rightarrow A_0$	-12.90	-12.90
A_2 coordination channel	$A_2 \rightarrow A_0, \Delta A_2 \bar{\xi}$	0	+5.90
Total	$\partial \mathbb{E}_t[\pi_{t+1}]$	-13.45	-7.87

- **Conventional term-premium channels are small** (< 1.2 bp): quantity and repricing nearly offset. This is the Corhay et al. (2023) accounting, and it is all a unique-equilibrium model sees.
- **The A_0 hedge channel is identical across equilibria** — it runs through $\lambda(\omega)$ and never touches A_2 . Alone, it makes QE disinflationary.
- **The A_2 coordination channel exists only in Eq II** and **halves** the disinflation, $-13.4 \rightarrow -7.9$ bp. Whether it *overturns* the sign is quantitative — see next slide.

When Does the Effect Reverse Sign?



- The gap between the curves *is* the coordination channel. It widens with duration because A_2^* does — so the two equilibria diverge most for long debt.
- **At baseline the effect is attenuated, not reversed.** A reversal needs $A_2^* \gtrsim 0.3$ against 0.21; that obtains for $\kappa \geq 0.2$ or $\nu \leq 0.2$. Over a grid of 320 parameter combinations, a reversal occurs in 62.
- **Honest scope:** a possibility the model admits over a sizable region, not a baseline prediction — and it turns on ν , the least identified parameter.

Does the Second Equilibrium Survive?

Modification	Survives?	At baseline	Why
Baseline	yes	$A_2^* = 0.206, \mu = -0.077$	Loop operative
Short-term debt, $\omega = 0$	yes	$A_2^* = 0.116, \mu = -0.095$	No duration amplification
Flexible prices, $\kappa \rightarrow \infty$	weaker	$A_2^* = 0.593, \mu \rightarrow 0$	CIR Jensen term, not the SDF loop
Active Taylor, $\rho_\pi > 1$	no	—	π recursion explosive, $PT < 0$

- Long maturity is *not* necessary. At $\omega = 0$ the second root survives. Duration amplifies the multiplicity but does not create it — unlike Aguiar–Amador (2020), where long bonds are indispensable.
- Sticky prices matter for the *mechanism*, not existence. As $\kappa \rightarrow \infty, \mu \rightarrow 0$ and the self-fulfilling SDF loop switches off — yet a second root persists, sustained purely by the CIR Jensen term $\frac{1}{2}k_1^2 A_2 v^2$. That flexible-price multiplicity is a *different*, weaker object.
- Active monetary policy destroys the equilibrium outright.

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What Coordinates Beliefs?

The model is silent on selection — both equilibria are fully rational, neither Pareto dominates. But the sunspot is *not* mystical: ξ_t is a **coordination state**, a public variable investors may or may not price.

Candidates that make Eq II more likely:

- Episodes that make the fiscal position *salient without changing it*: debt-ceiling standoffs, budget confrontations, rating actions, official warnings on sustainability, a narrative shift toward fiscal dominance.
- **Investor-base shifts**: from price-insensitive official holders toward leveraged / mark-to-market investors.
- **Safe-asset scarcity**: raises the marginal hedging value of government debt.
- **QE announcements themselves** — they make the maturity structure salient, which is exactly the margin governing how much the coordination channel is worth.

Policy feedback: the coordination channel is worth *more* when maturity is long, so lengthening maturity makes coordinating on Eq II more attractive. Endogenizing this (adaptive learning over μ ; local stability of the fixed points) is the natural next step.

Empirical Implications

Not estimated here, but the model gives sharp *sign* predictions.

- ① **Fiscal uncertainty moves IRP only in Eq II.** In Eq I, innovations to a fiscal-uncertainty proxy are orthogonal to inflation risk premia and real debt valuations. *Regime-dependence, not average strength, is the object.*
- ② **The sign of the bond response identifies the equilibrium.** Fiscal news raising bond prices and *lowering* inflation compensation = the deflation-hedge signature. Raising compensation and yields together = fiscal stress, outside this model.
- ③ **QE has state-dependent inflation effects** — a *sign flip* conditional on the pricing state, not a magnitude difference.
- ④ **Maturity interacts with fiscal-risk pricing:** since $\partial A_2^* / \partial \omega > 0$, valuations should load more on fiscal uncertainty where privately held maturity is longer.

Counterparts: TIPS breakevens and inflation swaps, decomposed IRP and term premia, fiscal uncertainty/news indices, privately held Treasury maturity, high-frequency QE surprises, survey dispersion (a natural proxy for the coordination state). Prediction 2 needs only a sign — the cheapest discriminating test.

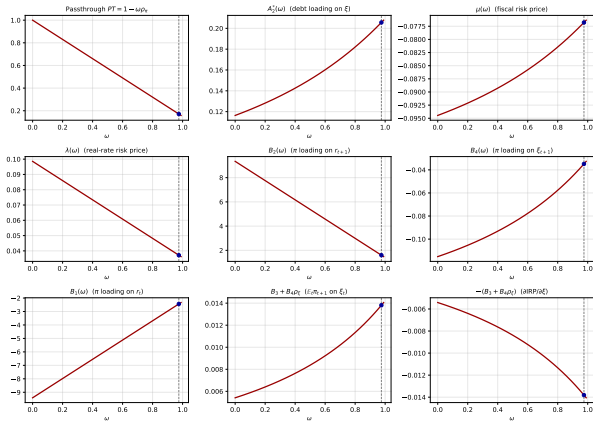
Conclusion

- The FTPL generates a novel indeterminacy: **not a multiplicity of price-level paths, but of inflation risk premia**. Within the affine Markov class the bond-pricing equation admits exactly two equilibria under identical fundamentals.
- They differ in *one* object — whether the pricing kernel loads on fiscal uncertainty. The self-validating equilibrium has a **negative** fiscal risk premium: government debt as insurance, the mirror image of the crisis literature's vicious loop.
- Debt maturity acts **asymmetrically**: the same passthrough factor PT dampens the fundamental channel and amplifies the self-fulfilling one.
- **Sharpest implication**: the effectiveness of debt maturity management is not a structural constant. The same operation is half as disinflationary when fiscal risk is priced, and reverses sign for a large enough fiscal loading — a candidate account of QE's ambiguous empirical record with *no* appeal to changing fundamentals.

Thank you

Appendix: Duration Comparative Statics

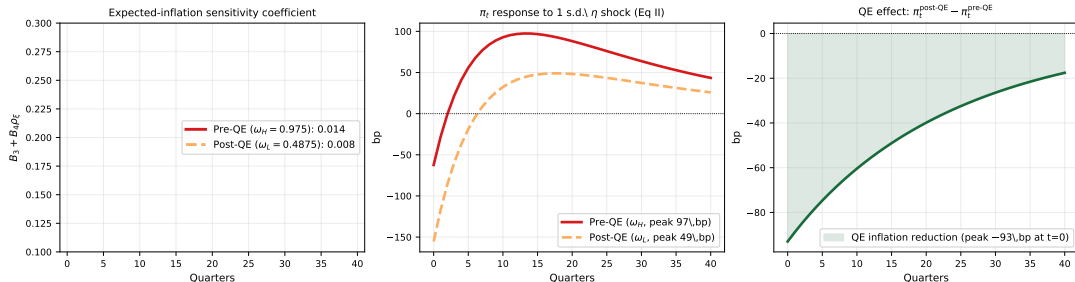
Sensitivity of equilibrium objects to bond duration ω (Fiscal Sunspot Equilibrium; baseline $\omega = 0.975$ dashed)



Each object over $\omega \in [0, 0.99]$; dashed line and dot mark the baseline. PT , (B_2, B_1) and λ shrink in magnitude with duration; A_2^* and $B_3 + B_4\rho\xi$ rise, so the safe-haven IRP deepens.

Appendix: Maturity Shortening Compresses the Sunspot Response

QE Effect on Fiscal Sunspot Inflation: Conditional Comparison
Pre-QE: $\omega_H = 0.975$, Post-QE: $\omega_L = 0.4875$ ($\rho_\omega = 0.9$, then reverts; shown at ω fixed)



$$\Delta = -0.006$$

Halving ω from 0.975 to 0.4875 lowers A_2^* and $B_3 + B_4\rho_\xi$, cutting the peak inflation response to a 1 s.d. η shock from +137 bp to +62 bp ($\approx -55\%$). This is the conditional-amplification effect, distinct from the level effect in the decomposition table.

Appendix: Four-Channel Decomposition, Derivation

Term-premium channels (Corhay et al. 2023) — from $\mathbb{E}_t[r_{g,t+1}] = i_t - \frac{1}{2}\text{var}_t + \gamma_t \text{TP}_t$:

$$\underbrace{\text{TP}_H \Delta\gamma_t}_{\text{(i) quantity}} + \underbrace{\gamma_t \Delta\text{TP}_t}_{\text{(ii) repricing}}$$

Small and offsetting: fewer units of duration, but each unit's premium rises as PT rises.

Balance-sheet channels — from the FTPL constant condition, where ω enters through both $\lambda(\omega)$ and $A_2(\omega)$:

$$\underbrace{-PT_t k_1 \Delta A_0^{(\lambda)}}_{\text{(iii-a) hedge, both eq.}} + \underbrace{-PT_t k_1 [(\Delta A_0 - \Delta A_0^{(\lambda)}) + \Delta A_2 \bar{\xi}]}_{\text{(iii-b) coordination, Eq II only}}$$

- $\Delta A_0^{(\lambda)}$ is computed in Eq I (where $A_2 \equiv 0$), isolating the pure $\lambda \rightarrow A_0$ route common to both equilibria.
- (iii-b) collects the $A_2 \rightarrow A_0$ shift *and* the direct ξ -loading repricing; both vanish when $A_2 \equiv 0$. Reversal needs (iii-b) > |(iii-a)|; at baseline +5.9 against -12.9, so it attenuates rather than overturns. Code: `src/inflme/qe.py`.